Ownership Change and Business-Model Reorientation in Nursing Homes*

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Abstract

Ownership changes in health care can reorganize what providers do, whom they serve, and how care is delivered. We study this process in U.S. nursing homes by identifying facility-level transfers of ownership control in CMS administrative data from 2017-2024 and tracing within-facility changes around those events. Ownership transfers are followed by substantial shifts in operating model: occupancy rises, Medicare share increases, Medicaid share falls, measured case mix increases, and average length of stay declines. Nursing labor also changes, with reductions in registered nurse and certified nursing assistant hours and little change in licensed practical nurse staffing. Quality responses are mixed. Some measures improve, including catheter use, urinary tract infections, hypnotic use, and short-stay pain, while preventive process measures such as influenza and pneumococcal vaccination deteriorate; several staffing-intensive resident outcomes show little average change. These patterns suggest that ownership transfers are not isolated governance changes applied to a stable care-production environment. Instead, they reveal broad business-model adjustment inside regulated health care providers, where changes in staffing and quality occur alongside simultaneous shifts in resident mix, payer composition, and throughout.

Keywords: nursing homes, ownership change, staffing, difference-in-differences

JEL Codes: I11, I18, J22, L22, L33

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1 Introduction

Ownership changes have become an increasingly common feature of the healthcare sector (footnote w/ citation?). Policymakers (Khan, 2024; Murphy and Scanlon, 2026; Oliveira and Blanco, 2024), academics (Stevenson et al., 2023; Ryskina et al., 2024; Gupta et al., 2024), and the public have expressed growing concern over the implications of these changes for patients and the quality of care. This concern is especially important in nursing homes as they serve an aging and medically vulnerable population who also face a wide range of frictions. Yet, existing literature gives limited or mixed results on how ownership changes affect patients, quality, and the facilities operating model. This is in part because ownership is difficult to measure and many studies only focus on specific forms of ownership, such as private equity, chain mergers, or for-profit acquisitions. This paper addresses that gap by introducing a new measure of verified substantive ownership changes and estimating their effects on both the production and quality of care in U.S. nursing homes.

This paper studies that question in U.S. nursing homes. We use CMS panel data from 2017 through 2024 to identify true transfers of economic control and estimate how staffing inputs and quality outcomes change following ownership changes. Nursing homes are a useful setting because prices are heavily shaped by public reimbursement, residents are medically vulnerable, and staffing is a central input into care production. If ownership changes affect care in this setting, the effects are likely to operate through internal production decisions: nurse staffing, staffing composition, task allocation, medication management, monitoring, documentation, and clinical routines.

The effect of ownership change is theoretically ambiguous because facilities may change owners for different reasons. Some transitions may reflect financial distress. A facility facing low occupancy, high labor costs, regulatory pressure, or declining profitability may be sold to an owner that expects to stabilize operations by reducing costs, tightening staffing, or standardizing management. In this case, ownership change may lead to input reductions, and whether those reductions harm residents depends on how close the facility was to the margin of adequate care. Other transitions may reflect succession or exit by incumbent owners. A small independent owner may sell because of retirement, limited access to capital, or limited ability to manage increasingly complex regulatory and administrative requirements. In these cases, a new owner may alter operations not because the facility is failing, but because the previous owner lacked the capacity or incentive to update practices.

A different set of ownership changes may reflect strategic acquisition or corporate restructuring. A chain may acquire a facility because it expects economies of scale in administration, purchasing, compliance, or labor management. A buyer may also acquire a facility because it believes existing operations can be reorganized without fundamentally changing the resident population or local market. Conversely, some ownership changes may be largely administrative or legal, producing little change in day-to-day care. These different motives imply different predictions: ownership change may generate quality-reducing cost containment, little operational change, or efficiency-enhancing reorganization. Which of these responses occurs is ultimately an empirical question.

Distinguishing among these possibilities requires examining both inputs and outcomes. Staffing reductions alone are difficult to interpret. They may reflect quality-reducing cost cutting, but they may also reflect the removal of inefficient routines or changes in the organization of care. Quality measures alone are also incomplete, because observed improvements may occur alongside reductions in labor inputs that matter for unmeasured aspects of resident welfare. The joint response of staffing and quality is therefore central. If ownership changes primarily reduce inputs in ways that harm residents, staffing declines should be accompanied by worsening in resident outcomes. If staffing falls while quality measures improve or remain stable, the pattern is more consistent with reorganization of production. In that case, ownership change may reveal that some facilities had previously operated inside their feasible production frontier: not because new owners have a superior technology, but because changes in control can disrupt routines, alter managerial attention, and change how labor is allocated across tasks.

Ownership changes are not random events, which creates an important empirical challenge. Facilities that change owners may differ from those that do not, and the conditions leading to a sale may also shape subsequent staffing and quality. Our analysis therefore focuses on within-facility changes around transfers of control rather than cross-sectional differences between facilities with different owners. In a staggered difference-in-differences event-study design, facilities are compared to themselves before and after ownership change, relative to contemporaneous changes among facilities that have not yet changed ownership or do not change ownership during the sample period. This design does not eliminate all concerns about endogenous timing, but it allows us to ask a more specific question: how do staffing and quality evolve when control of an existing facility changes hands?

This paper contributes to a growing literature on ownership, consolidation, and quality in health care. Prior work emphasizes that mergers and ownership changes can affect provider behavior through several channels, including market power, economies of scale, financial incentives, organizational restructuring, and changes in workforce composition. Much of the existing evidence focuses on hospital mergers, physician-practice consolidation, private equity acquisitions, or differences across ownership forms. These studies show that the effects of ownership and consolidation on quality are often mixed and context-dependent. Some mergers appear to reduce capacity without improving productivity or quality; others improve management or quality measures while worsening patient experience or leaving broader clinical outcomes unchanged. This evidence suggests that ownership changes should not be understood only as changes in market structure. They may also operate through internal production decisions.

Nursing homes are a particularly useful setting for studying these internal production channels. The sector is labor intensive, heavily regulated, and financed largely through public reimbursement. Because prices are constrained, staffing and care routines are natural margins of adjustment. Nursing home ownership has also been a persistent policy concern. Prior work finds that ownership form, chain affiliation, private equity ownership, and public reporting can affect staffing, quality, and resident outcomes. However, less is known about what happens when control of an existing nursing home changes hands while the facility itself remains in the same market. This distinction matters. A transfer of control does not necessarily alter the local set of providers, but it may change the decision-maker responsible for staffing levels, staffing mix, monitoring systems, medication management, documentation, and other care routines.

Nursing labor is the core input in nursing home care production, and staffing is one of the most consequential operation choices facilities make. We distinguish among registered nurses (RNs), licensed practical nurses (LPNs) and certified nursing assistants (CNAs). RNs include registered nurses and directors of nursing and play central supervisory and clinical roles, including assessing residents' conditions, developing and coordinating care plans, overseeing medication-related processes, and supervising other staff. LPNs provide routine clinical services and medication management tasks such as monitoring vital signs, while CNAs deliver the majority of hands-on daily care, assisting with bathing, dressing, toileting, feeding, and mobility. Because these staff types differ in training, scope of practice, and cost, ownership changes may affect not only total staffing hours but also the skill mix of care delivered. For example, an owner seeking quick cost containment may reduce total hours or substitute away from higher-cost licensed nurses, while an owner focused on compliance or clinical quality may increase RN oversight or adjust supervision structures.

Our results show that ownership changes are followed by sustained reductions in nursing labor inputs. RN hours and CNA hours decline, and total nursing hours fall. LPN hours remain largely unchanged. These staffing responses suggest that ownership transitions alter both the level and composition of labor used in care production. Our quality results complicate a simple cost-cutting interpretation. Labor saving mechanism measures, including catheter use and anti-anxiety or hypnotic medication use, improve after ownership change. Urinary tract infections also decline. By contrast, resident outcome measures such as pressure injuries, falls with major injury, weight loss, and decline in physical functioning show little evidence of broad worsening.

The paper makes three contributions. First, it contributes to the literature on health care ownership and consolidation by studying ownership change as a shock to internal production rather than only as a change in market structure. Second, it contributes to the nursing home literature by jointly examining staffing composition and measured quality outcomes following verified transfers of economic control. Third, it contributes to broader work on organizational change by showing that changes in governance can disrupt routines and alter production even when technology, location, regulation, and the local provider set remain largely fixed.

The following sections outline the rest of this paper. Section 2 provides background on the nursing home industry and ownership structures. Section 3 describes the data, ownership-change measurement, staffing and quality outcomes, and empirical strategy. Section 4 presents the results. Section 5 discusses interpretation and limitations. Section 6 concludes.

2 Institutional Background

2.1 The Change-of-Ownership Process

The date on which a nursing home changes hands in administrative data is the end point of a process that begins months earlier, and the gap between the two is central to our identification strategy. A sale is negotiated privately, and the parties sign a binding asset purchase agreement or stock purchase agreement well before the transfer takes legal effect. Closing typically follows signing by several months, with the interval usually determined by state licensure review: the incoming licensee must be approved by the state survey agency, and in some states must also clear certificate-of-need or character-and-fitness review. Requirements and processing times vary substantially across states. During this signing to closing window the buyer frequently acquires contractual influence over operating decisions, while the seller has a strong incentive to present stable census and clean financials. Operational changes, including staffing decisions, can therefore begin before the recorded ownership-change date.

On the federal side, the incoming owner files a Form 855A change-of-ownership application and elects either to accept automatic assignment of the seller's existing Medicare provider agreement or to reject it. Accepting assignment preserves the facility's certification and its CMS Certification Number (CCN), but carries forward its deficiency history and successor liability for prior overpayments; rejecting assignment requires survey as an initial applicant and creates a gap in Medicare participation. Most buyers accept assignment. This matters for our design because the CCN persists across the transfer, the facility remains a continuously identifiable unit in administrative data before and after the change of ownership, which is what permits a within-facility comparison around the event.

The ownership-change date observed in HCRIS is the reported effective date of the transfer.[1] The measurement concern is therefore not that the recorded date lags the transfer, but rather, the operational adjustment precedes it. As a result, the months immediately before the effective date may already reflect responses to the pending sale rather than untreated facility behavior. Section 3 describes the resulting manner in which we handle this.

2.2 The Reimbursement Environment

Nursing home revenue is determined overwhelmingly by public payers at administered prices. Medicaid is the dominant payer for long-stay residents, who account for the majority of resident-days. Rates are set at the state level, vary in whether they are cost-based or price-based, and are generally low relative to the cost of care. Medicare Part A covers short-term post-acute stays following a qualifying hospitalization, subject to a benefit cap of 100 days per spell of illness and, historically, a three-day prior inpatient stay requirement that was waived during the COVID-19 pandemic. Since October 2019, Medicare has paid skilled nursing facilities under the Patient-Driven Payment Model, which replaced the Resource Utilization

[1] CMS separately issues a "tie-in notice" recording the transfer in its enrollment systems, which may follow the effective date by several months. Because we use the reported effective date, this administrative processing lag does not shift our treatment date.

Group system used earlier in our sample and changed how resident clinical characteristics map into payment.

The feature of this environment that matters for our analysis is the gap between the two payers: Medicare per-diem payments are substantially higher than Medicaid per-diem payments for an otherwise comparable resident-day. (I will cite the actual ratio). Because a facility cannot adjust price, the margins available to an owner seeking to improve financial performance are the composition of the resident census, the intensity and composition of labor inputs, and throughput. Raising the Medicare share of patient days, filling empty beds, and shortening length of stay are all revenue responses to a payment environment in which price is fixed. This is why we treat occupancy, payer mix, and length of stay in Section 4 as outcomes rather than as descriptive controls.

2.3 Nursing Labor and Staffing Regulation

Nursing labor a facilities largest operating cost, and the three staff types we observe are imperfect substitutes. Registered nurses (RNs), including directors of nursing (ASK ABOUT THIS), perform resident assessment, care planning, MDS coordination, infection surveillance, and supervision of licensed staff. Licensed practical nurses (LPNs) administer medications and perform treatments under supervision. Certified nursing assistants (CNAs) provide assistance with activities of daily living and account for the large majority of direct daily care. Substitution among them is legally constrained: tasks within the RN scope of practice cannot be reassigned to CNAs, so a facility cannot replace RN hours with CNA hours and deliver the same care. Compensation follows similarly, with an RN hour costing roughly twice a CNA hour. (I can cite the actual BLS numbers here if it matters).

Staffing is bounded by regulation, although the limits are looser than some may think. Federal rules require nursing staff sufficient to meet resident needs, an RN on duty at least eight consecutive hours per day seven days per week, a licensed nurse present twenty-four hours per day, and a full-time RN serving as director of nursing. State minimum-staffing laws operate alongside these requirements and vary considerably in stringency and enforcement. CMS finalized a national minimum staffing rule in 2024 with phase-in scheduled to begin after our sample ends, and that rule was subsequently repealed; neither binds during the period we study.

These features together identify RN hours as the most available margin to a cost-conscious acquirer. The federal floor for RN coverage is thin relative to other staffing, so RN hours contain slack that regulation does not protect. RN functions are disproportionately administrative, preventive, and supervisory, making reductions less immediately visible to residents, families, and surveyors. And because RN hours are the most expensive, each hour removed yields the largest cost saving. The combination of regulatory slack, low observability, and high unit cost makes RN staffing the natural first margin of adjustment.

Finally, the nursing labor market is unusually unstable, which shapes staffing adjustments. Turnover is high across all three staff types and especially among CNAs, and facilities rely heavily on contract and agency staffing to maintain coverage. These pressures intensified during the COVID-19 pandemic, when facilities faced increases in acuity alongside a contracting pool of available workers. An acquirer's ability to reduce staffing therefore depends on local labor market conditions at the time of the transaction, which motivates the pre- and post-pandemic comparison reported below.

2.4 Ownership Churn and Its Measurement

Ownership in the sector is highly disconnected, with a large number of small independent operators facing succession pressure, limited access to capital, and rising administrative burden. Thin operating margins and the separation of real estate from operations have also made facilities attractive to real estate investment trusts and private equity investors. In our sample roughly one in five experience a verified ownership change over the seven and a half year panel.

Measuring these changes is not straightforward, which motivates our approach described in Section 3. Ownership structures are frequently layered, with an operating entity held by other corporations, trusts, or partnerships, and a change in the listed owner does not necessarily correspond to a change in who actually controls the facility. Legal restructuring within a single ownership group, transfers among family members, and changes in reporting conventions can all appear as ownership changes in administrative records without any transfer of operating control. Distinguishing substantive transfers from these administrative events is a central measurement problem for this literature and a primary contribution of this paper.

3 Data and Methods

3.1 Data Sources

We utilize three main sources of data: Nursing Home Compare (NHC) Archive data, Healthcare Cost Report Information System (HCRIS), and Payroll Based Journal (PBJ) data. The NHC Archive contains monthly facility level information on ownership, chain affiliation, certification status, as well as quarterly reported quality measures. Second, we use HCRIS, also known as the Medicare Cost Reports, which contains annual cost reports submitted by every Medicare certified nursing home in the United States. The cost reports contain information about costs, revenue, financial data for each reporting facility, and dates of ownership changes. Lastly, we use the PBJ data to obtain staffing information. The PBJ data contains the resident census and the number of hours worked by various types of NH staff on a daily basis.

The three sources differ in how and when they are reported, which has consequences for our analysis. PBJ is reported daily and aggregated to facility-months, so staffing outcomes vary at monthly frequency. NHC quality measures are reported quarterly. HCRIS, by contrast, is reported once per cost-reporting fiscal year. We expand each cost report across the months spanned by its fiscal year, assigning the reported fiscal-year value to every month in that period. Facility characteristics drawn from HCRIS (occupancy, payer mix, average length of stay, spare capacity, and certified beds) therefore change at most once per fiscal year and are constant within a facility's fiscal year by construction. For quarterly specifications, HCRIS-derived characteristics are aggregated from months to quarters using that quarters mode. [2]

3.2 Sample selection

We utilize two samples in our study, the first is our staffing panel which is constructed at the facility-month level and the second is our quality metric panel which is constructed at the facility-quarter level. We construct both samples using the following criteria. First, we limit our samples to the 48 contiguous U.S. states from the period January 2017 to June 2024. Secondly, the NHC Archive data is matched to the HCRIS data on a unique nursing home provider number. Because HCRIS only contains information on freestanding nursing homes, our samples exclude hospital-based nursing homes. Hospital based nursing homes differ from freestanding facilities in their structure and financing. Because they are within hospitals, staffing policies and resource allocation may be jointly determined with hospital operations and respond to hospital shocks and regulations rather than nursing home specific incentives. We therefore exclude hospital based facilities to maintain a more homogeneous set of providers and to ensure ownership changes reflect changes in nursing home management rather than broader hospital system changes. Hospital based nursing homes make up about 12 percent of all nursing homes in the United States.

We also exclude facilities that at any time during our sample were government owned. In some states, facilities that are privately owned are able to change their legal ownership to government entities while still keeping control of a facility. These arrangements, which are concentrated in a small number of states, allow a facility to qualify for supplemental Medicaid payments available to publicly owned providers while day-to-day operations, man-

[2]If there is a tie, e.g., a quarter with a month of no data and two months where the data changed, then we use the most recently observed month.

agement, and staffing remain with the original operator. (I can cite the actual example we encountered). Because these transfers are reimbursement driven rather than substantive transfers of operating control, treating them as ownership changes would misclassify a purely financial arrangement as an operational one. Rather than attempt to separate these cases from genuine transfers involving government entities, we exclude any facility observed as government-owned at any point in the panel. The same exclusion is applied to the quality sample so that both panels are estimated on a common set of facilities.

3.3 Identifying Ownership Changes

No single source provides a complete record of substantive ownership changes in U.S. nursing homes. We therefore construct a facility-level dataset of ownership changes intended to capture transfers of substantive economic control, rather than purely legal or administrative changes. Existing studies of nursing home ownership changes and mergers generally rely on two primary sources: HCRIS and the NHC Archive. Each source contains useful information, but each is incomplete when used alone. We therefore use a two-step identification and verification process that combines reported ownership-changes in HCRIS with monthly ownership records in the NHC Archive.

HCRIS provides a useful starting point because it records whether a facility reported a change in ownership during the reporting period and, when applicable, the reported date of the change. These records identify ownership changes that facilities report through the cost-reporting process. However, HCRIS does not by itself establish whether a reported change reflects a substantive transfer of economic control. Some HCRIS-reported changes may capture legal restructuring, administrative reorganization, or changes in ownership where decision-making authority remains within the same ownership group. As a result, relying on HCRIS alone may overstate the number of ownership changes that are economically meaningful. Appendix Table **??** provides examples of HCRIS-reported ownership changes that are not classified as substantive transfers, and therefore why HCRIS-reported changes need verification.

The NHC Archive provides more granular ownership information. It reports monthly owner roles, owner types, owner names, and ownership percentages for each facility. This allows us to examine whether the individuals or entities controlling a facility change over time. However, the detail in the NHC Archive also introduces measurement challenges. Nursing home ownership structures are often complex and layered. For example, a limited liability company may directly own a nursing home, while the company itself is owned by other individuals, corporations, trusts, or partnership entities. Changes in listed owners, ownership layers, or reporting conventions can therefore appear as ownership changes even when economic control has not clearly changed. Programming comparisons of monthly NHC ownership records can consequently generate false positives if used without additional validation. Appendix Table **??** also provides examples in which NHC ownership records indicate large changes in listed owners or ownership shares but no corresponding HCRIS ownership-change flag is observed.

We combine the two sources to address these limitations. HCRIS provides reported ownership changes and dates, while the NHC Archive allows us to verify whether those events correspond to changes in the underlying ownership structure. In constructing the NHC Archive ownership-change measure, we identify ownership changes at the highest ownership level observed for a facility. Specifically, when multiple ownership layers are present, we prioritize indirect owners, then direct owners, then partnership-interest owners. This is intended to capture changes in economic control at the highest available ownership layer rather than changes in lower-level legal entities that may not correspond to a change in decision-making authority.

We define a substantive ownership change as a case in which a majority (i.e. greater than 50%) of ownership shares turns over between month $t-1$ and month t. This is intended to capture transitions in which control of the facility changes hands, including cases where a new owner acquires most or all ownership shares. When ownership percentages are partially or completely missing, we normalize any reported percentages and, where necessary, assign equal weights across remaining owners. To reduce false positives from legal restructuring within the same family ownership group, we do not flag a change when at least 80 percent of the ownership stake remains within the same family.

Using these verified ownership changes, our final samples include nursing homes that either have no ownership change in either source or have exactly one ownership change in both sources, with the HCRIS and NHC ownership-change dates occur within six months of one another. For treated facilities, we define the ownership-change month as the month in which HCRIS identifies the change. All dates are converted to calendar months, so changes occurring at the beginning or end of a month are treated identically. Appendix Table **??** summarizes the ownership-change classification process and the construction of the ownership-change sample.

3.4 Nursing Staff Levels

We measure staffing separately for the three nurse categories described in Section 2.3: RNs, LPNS, and CNAs. Because these categories differ in training, scope of practice, and cost, changes in staffing composition may reflect different organizational responses to ownership change than changes in total staffing alone, and we therefore report each category separately as well as a total.

Staffing is measured from PBJ, which reports hours worked by staff category and a daily resident census drawn from MDS assessments. Reported hours include both employee and contract staff.

For each facility i, calendar month t, and staff type, we aggregate daily hours and resident days to construct hours per resident day (HPRD):

$$\text{HPRD}_{it} = \frac{\sum\limits_{d \in t} H_{id}}{\sum\limits_{d \in t} R_{id}}, \tag{1}$$

where H_{id} is the number of hours worked by the staff type in facility i on day d, and R_{id} is the number of resident days in facility i on day d. We construct separate HPRD measures for RNs, LPNs, and CNAs, and define total HPRD as the sum of the three type-specific measures. For quality specifications estimated at the facility-quarter level, we construct quarterly staffing measures analogously by summing hours and resident days over all days in the quarter before taking the ratio.

We also report the numerator of this ratio directly. Raw monthly hours, $\sum_{d \in t} H_{id}$, measure the total quantity of nursing labor purchased by the facility without normalizing by census. Reporting both is necessary because the HPRD denominator is itself an outcome in this paper: if occupancy rises following an ownership change, HPRD can fall even when the facility has not reduced the hours it employs. Separating hours from HPRD allows us to distinguish a reduction in labor purchased from a consequence of a larger resident census. We report raw hours in levels and logs alongside the corresponding HPRD measures.

In addition to nursing staff, PBJ reports hours for therapy and other non-nursing personnel. We construct HPRD measures for physical therapists, physical therapy assistants and aides, occupational therapists, occupational therapy assistants and aides, and speech-language pathologists, along with a total non-nurse measure. These are treated as secondary outcomes. We report them in levels only: several categories, particularly therapy aides, contain a large share of exact zeros which makes log specifications less informative than for nursing staff.

We also construct additional measures of data quality. First, we calculate a monthly PBJ coverage measure, defined as the share of days in the month with complete staffing records. Secondly, we track reporting gaps, defined as the number of months since a facility's previous PBJ observation, and use this variable in robustness checks that restrict the sample to facilities with more continuous reporting.

Finally, we exclude a small number of observations with implausible staffing levels: months in which RN and LPN HPRD are both zero, total HPRD is below 1.5 or above

12, or CNA HPRD exceeds 5.25. These observations are outliers that likely reflect reporting errors rather than genuine variation in staffing. These exclusion criteria align with the criteria outlined in the CMS NHC Technical Users Guide.

3.5 Quality Metrics

We measure quality using selected CMS quality measures from the NHC Archive. These measures are reported at the facility-quarter level and are constructed from resident assessments submitted through the Minimum Data Set (MDS). We focus on measures that are plausibly responsive to ownership-related changes in staffing, care routines, medication management, monitoring, and documentation. For each measure used in the analysis, lower values indicate better measured quality.

We group the selected measures into two categories. The first category consists of labor saving mechanism measures: catheter use, anti-psychotic medication use, and anti-anxiety or hypnotic medication use. Catheter use measures the percentage of long-stay residents with a catheter inserted and left in their bladder. Catheters may be clinically appropriate in some cases, such as urinary retention, selected wound-care situations, or end-of-life care, but unnecessary catheter use can increase infection risk and may reflect facility routines around toileting assistance and monitoring. Anti-psychotic medication use measures the percentage of long-stay residents who received an anti-psychotic medication and anti-anxiety or hypnotic medication use measures the percentage of long-stay residents who received an anti-anxiety or hypnotic medication. Both of these metrics capture medication-management practices among long-stay residents. These medications may be clinically appropriate for some residents, but their use can also reflect prescribing routines, behavioral-management practices, staff monitoring, and reliance on pharmacological approaches to resident care.

The second category consists of resident outcome measures: pressure injuries, falls with major injury, weight loss, decline in physical functioning, and urinary tract infections. These measures are more directly related to resident health and functioning and may be affected by staffing-intensive care processes. Pressure injuries may reflect repositioning, skin checks, hygiene, nutrition, and monitoring. Falls with major injury capture serious resident safety events that may be affected by supervision, mobility assistance, medication management, and environmental safety. Weight loss captures clinically meaningful unintended weight loss [3] and may reflect nutrition, feeding assistance, monitoring, or changes in resident condition. Decline in physical functioning captures worsening dependence in mobility or activities of daily living. Urinary tract infections may reflect resident frailty and clinical need, but also catheter use, hydration, toileting assistance, hygiene, monitoring, and documentation practices.

The measures above are constructed from assessments of long-stay residents. We additionally report two short-stay measures: the percentage of short-stay residents who newly received an antipsychotic medication, and the percentage of short-stay residents whose physical function improved. Short-stay measures are constructed from a different resident population than the long-stay measures, so estimates for the two groups are not directly comparable to one another and are reported in separate panels.

Several measures are not reported continuously across the sample period. CMS discontinued or substantially revised the reporting of some measures during our window, producing intervals in which a measure is largely or entirely unreported. Where this occurs we restrict estimation for that measure to the years in which it is reported: pressure injuries are estimated on 2018–2023 and short-stay improvement in function on 2017–2022. These restrictions apply to the individual measure only and do not affect the sample used for other outcomes.

3.6 Business-Model Outcomes

We examine a set of facility characteristics that describe the operating model rather than the production of care directly. These outcomes capture whom the facility serves, how full it

[3] This does not include residents who are on a physician-prescribed weight-loss plan.

is, and how quickly residents turn over, and they are the principal revenue margins available to an owner facing administratively determined prices, as described in Section 2.2. All are constructed from HCRIS and, as noted above, vary at most once per fiscal year.

Occupancy is the share of available bed-days that are occupied, computed as total patient days divided by available bed-days and expressed as a percentage. When available bed-days are not reported, we use certified beds multiplied by the number of days in the fiscal year as the denominator. Spare capacity is the share of certified beds that are unoccupied on an average day, computed as certified beds minus average daily census, divided by certified beds, where average daily census is total patient days divided by the number of days in the fiscal year.

Payer mix is measured as the share of total patient days attributable to Medicare and to Medicaid respectively. We note that these are shares of patient days rather than shares of revenue; because Medicare and Medicaid per-diem rates differ substantially, a given shift in the patient-day mix corresponds to a larger shift in the revenue mix. Average length of stay is taken directly from the cost report. Certified beds is the facility's total certified bed count.

We also construct a measure of resident acuity from the case-mix index reported in the NHC Archive. CMS altered the methodology used to construct case-mix indices during our study period, so raw case-mix values are not comparable across the full window. Where case mix is used as a control we address this by measuring acuity in relative rather than absolute terms, constructing case-mix quartiles within each state and calendar month.

3.7 Control Specifications

A central difficulty in choosing controls for this setting is that ownership change plausibly affects many of the facility characteristics that would conventionally be included as covariates. Occupancy, payer mix, average length of stay, and resident acuity are themselves outcomes in this paper. Conditioning on them when estimating effects on staffing or quality would absorb part of the response we are attempting to measure, and would bias estimates toward zero in exactly the dimension of interest. Rather than adopt a single control set, we organize covariates into four nested specifications ordered by how plausibly exogenous each regressor is with respect to the transfer of ownership.

Specification A includes the facility's certified bed count and its baseline chain affiliation. Bed count is the facility's licensed capacity, which is costly to adjust and changes infrequently. Baseline chain affiliation is the facility's chain status measured at the start of its panel and held fixed thereafter, rather than the contemporaneous chain indicator. We use the baseline measure because chain status is self-reported and changes around ownership transactions themselves, making the time-varying indicator endogenous to treatment. Because baseline chain affiliation is time-invariant, it is absorbed by the facility fixed effects and does not appear separately in estimated coefficients; it is retained in the specification definition because it is used to define subsamples in the heterogeneity analysis.

Specification B adds non-profit status and case-mix quartile indicators. Specification C further adds occupancy, Medicare and Medicaid shares of patient days, and average length of stay. Specification D further adds RN, LPN, and CNA hours per resident day. Each tier is applied only where it is well defined: Specification B is not used when case mix or non-profit status is itself the outcome, Specification C is not used for business-model outcomes because those variables constitute the business-model outcome set, and Specification D applies only to quality outcomes.

Specification A is our preferred specification and is used for all main results reported in Section 4. Specifications B through D are reported separately as a decomposition exercise, showing how the estimated effect moves as progressively more treatment-responsive regressors are added to the model. We interpret the movement across tiers as informative about the channels through which ownership change operates, not as a sequence of increasingly credible estimates.

Revise: The event-study figures currently in the paper were generated under the older full control set rather than Specification A. They must be regenerated before this description is accurate; twfe_event_study.R and quality_event_study.R both still call make_controls_rhs().

Two variables that appear in related work are deliberately absent. Ownership type does not enter as a control: government-owned facilities are excluded from the sample entirely, so that indicator is constant, and for-profit status is the omitted reference category against which non-profit status is measured in Specification B and above. The contemporaneous chain indicator likewise does not enter any specification, for the endogeneity reason described above.

Revise: Chain status is derived from the HCRIS home-office field rather than NHC, and facilities with a missing home-office entry are coded as non-chain rather than missing. State this explicitly and confirm it is the intended treatment of missing values.

3.8 Analytic Approach

Ownership changes occur at different dates across facilities during the study period. Because these changes are not randomly assigned, our empirical strategy does not compare facilities that change ownership to facilities that do not in levels. Instead, we exploit the timing of verified control-changing ownership transitions to compare changes within a facility before and after ownership change, relative to contemporaneous changes among facilities that have not yet experienced a change or never experience one during the study period. All specifications include facility fixed effects, calendar-period fixed effects, and the covariates defined by Specification A.

This staggered timing raises a standard concern for two-way fixed effects designs. In staggered-adoption settings, conventional two-way fixed effects estimates can be sensitive to treatment-effect heterogeneity and comparisons involving already-treated units (Goodman-Bacon, 2021). We therefore emphasize event-study estimates using alternative comparison groups. This design is well suited to the institutional setting. Following an ownership change, the facility remains in the same location, operates under the same certification framework, and serves the same local market. As a result, any post-transition changes are likely to operate through internal production decisions, including staffing, routines, and management practices, rather than through a mechanical change in local market structure.

A complication in defining event time is that the recorded ownership change month may not coincide with the beginning of the transfer process. The ownership change date we observe in HCRIS is the effective date of the change, or the closing date. However, the sale process begins earlier, when the buyer and seller negotiate and sign a contract. Ideally, we would observe the contract signing date or the month in which transition planning begins. Because these dates are not observed, the months immediately preceding the recorded closing date may already reflect operational responses to the pending transfer. During this pre-transfer period, incumbent managers or prospective owners may adjust staffing policies, scheduling, hiring, agency use, or other operating routines. Because staffing can adjust relatively quickly, these months may not represent untreated facility behavior. For this reason, the preferred staffing event-study specification excludes $\tau = -3, -2, -1$ and uses $\tau = -4$ as the reference period. For quality outcomes, which are measured quarterly, the timing concern is different. The ownership-change quarter, $\tau = 0$, may contain care, assessment, and documentation from both before and after the transfer. We therefore exclude $\tau = 0$ and use $\tau = -1$ as the reference period.

We estimate the following set of event-study regressions:

$$S_{im} = \sum_{\tau=-24,\tau\neq-4,-3,-2,-1}^{24} \beta_\tau \mathbf{1}[T_{im} = \tau] + X_{im}\gamma + \mu_i + \lambda_m + \varepsilon_{im}. \tag{2}$$

For staffing outcomes, S_{im} denotes RN, LPN, CNA, or total HPRD for facility i in month m, or the corresponding raw-hours measure. The variables $\mathbf{1}(T_{im} = \tau)$ are event-time indicators for whether facility i is τ months from ownership change. The coefficients β_τ trace monthly staffing dynamics relative to four months before the ownership change. Two distinct operations are reflected in the summation index. The period $\tau = -4$ is the omitted reference category, against which all other coefficients are measured. The periods $\tau = -3, -2, -1$ are excluded in a different sense: facility-months falling in this window are dropped from the estimation sample entirely, so no coefficient is estimated for them. This removes the interval most likely to be contaminated by the pre-transfer adjustment described in Section 2.1.

$$Q_{iq} = \sum_{\tau=-8,\tau\neq-1,0}^{8} \theta_\tau \mathbf{1}[T_{iq} = \tau] + X_{iq}\pi + \mu_i + \lambda_q + \nu_{iq}. \tag{3}$$

For quality outcomes, Q_{iq} denotes a selected CMS quality measure for facility i in quarter q. The variables $\mathbf{1}(T_{iq} = \tau)$ are event-time indicators for whether facility i is τ quarters from ownership change. The coefficients θ_τ trace quarterly quality dynamics relative to the quarter before ownership change. The ownership-change quarter, $\tau = 0$, is excluded because it may contain both pre- and post-transfer care, documentation, and reporting.

In both equations, X_{im} and X_{iq} denote the Specification A covariate vector described above. Because baseline chain affiliation is time-invariant and absorbed by the facility fixed effects, the certified bed count is the only covariate that enters with an estimated coefficient. Facility fixed effects μ_i absorb time-invariant differences across nursing homes, while month fixed effects λ_m and quarter fixed effects λ_q absorb common shocks over time. Standard errors are two-way clustered by facility and calendar period.

For quality outcomes, we estimate specifications both with and without RN, LPN, and CNA HPRD controls. Because staffing is itself affected by ownership change, the staffing-controlled specifications are not interpreted as preferred estimates of the total effect of ownership change on quality. Instead, they are used descriptively to assess whether the estimated quality responses are attenuated after conditioning on measured staffing inputs.

We also estimate average post-change specifications that replace the event-time indicators with a single post-ownership-change indicator:

$$Z_{ip} = \delta Post_{ip} + X_{ip}\gamma + \mu_i + \lambda_p + \varepsilon_{ip}. \tag{4}$$

In equation (4) $p = m$ for staffing outcomes and $p = q$ for quality outcomes. Z_{ip} denotes either a staffing outcome in facility-month im or a quality outcome in facility-quarter iq. The indicator $Post_{ip}$ equals one after the ownership change and zero otherwise. The coefficient δ summarizes the average post-ownership-change difference in the outcome, relative to the pre-change period, after accounting for facility fixed effects, time fixed effects, and time-varying controls. For staffing outcomes, the preferred post specifications are estimated after excluding the same immediate pre-transfer months used in the event-study specification. For quality outcomes, the preferred post specifications exclude the ownership-change quarter. These post specifications are used to summarize the average post-change difference and should be interpreted alongside the event-study estimates.

For staffing outcomes, we estimate event study and conventional difference-in-differences in both levels and logs. Log specifications allow coefficients to be interpreted as approximate percentage changes and reduce sensitivity to differences in baseline staffing levels across facilities. Log specifications are omitted when staffing is zero.

3.8.1 Parallel Trends

The identifying assumption of equations 2 and 3 is, absent an ownership change, treated facilities would have followed similar staffing and quality trends as not-yet-treated and untreated facilities. We assess this assumption using the pre-treatment coefficients in the event-study models and joint Wald tests of whether these coefficients are equal to zero.

For staffing outcomes, we compare pre-trend tests from the full pre-window to those from the preferred donut specification. The full pre-window includes the months immediately preceding the recorded ownership change and uses $\tau = -1$ as the reference period. The donut specification excludes $\tau \in \{-3, -2, -1\}$ and uses $\tau = -4$ as the reference period. Table 1 reports the results of these Wald tests. The improvement in the pre-trend tests after removing the immediate pre-transition months supports the interpretation that those months are influenced by the pre-transfer adjustment window.

Table 1: Joint Wald Tests of Pre-trends for Monthly Staffing

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
Panel A: HPRD				
2 Year Full Pre-Window	93.04 (23)	39.36 (23)	36.25 (23)	48.73 (23)
	[0.0000]	[0.0181]	[0.0389]	[0.0013]
2 Year Window with Donut	27.52 (20)	34.10 (20)	11.36 (20)	23.90 (20)
	[0.1213]	[0.0254]	[0.9362]	[0.2466]
1 Year Window with Donut	11.23 (8)	6.39 (8)	1.43 (8)	3.73 (8)
	[0.1892]	[0.6039]	[0.9938]	[0.8806]
Panel B: Log(HPRD)				
2 Year Full Pre-Window	47.06 (23)	38.05 (23)	35.80 (23)	44.39 (23)
	[0.0022]	[0.0252]	[0.0433]	[0.0047]
2 Year Window with Donut	26.11 (20)	37.18 (20)	10.86 (20)	22.71 (20)
	[0.1622]	[0.0111]	[0.9498]	[0.3033]
1 Year Window with Donut	4.30 (8)	6.81 (8)	1.74 (8)	4.45 (8)
	[0.8294]	[0.5574]	[0.9879]	[0.8140]

Notes: Each cell reports the Wald χ^2 statistic for the joint null that all pre-treatment event-time coefficients equal zero, followed by degrees of freedom in parentheses and the p-value in brackets.

Tested windows and reference periods: 2 Year Full Pre-Window tests $\tau = -24$ to $\tau = -2$ with reference $\tau = -1$; 2 Year Window with Donut tests $\tau = -24$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$); 1 Year Window with Donut tests $\tau = -12$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$).

Sample sizes (rows): 2 Year Full Pre-Window ($N = 630,420$); 2 Year Window with Donut ($N = 626,311$); 1 Year Window with Donut ($N = 626,311$).

All specifications include facility and month fixed effects and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.

For quality outcomes, we conduct analogous Wald tests using the quarterly event-study specification. The full pre-window tests whether the coefficients for quarters $\tau = -8$ through $\tau = -1$ are jointly equal to zero. Our preferred quality specification excludes the ownership-change quarter, $\tau = 0$, because this quarter may combine pre- and post-transfer care, assessment, and documentation. Table 2 reports the results of these Wald tests.

Table 2: Joint Wald Tests of Pre-Trends for Quarterly Quality Measures

Outcome	**2 Year Full Pre-Window**	**2 Year, with Donut**	**1 Year, with Donut**
Catheter	12.10 (7) [0.0972]	13.00 (7) [0.0721]	2.01 (3) [0.5707]
Antipsychotic	7.49 (7) [0.3793]	7.89 (7) [0.3425]	3.43 (3) [0.3301]
Hypnotics	3.29 (7) [0.8570]	3.33 (7) [0.8532]	0.53 (3) [0.9117]
Pressure injuries	12.76 (7) [0.0781]	12.62 (7) [0.0820]	1.77 (3) [0.6211]
Falls	4.13 (7) [0.7649]	3.88 (7) [0.7937]	0.24 (3) [0.9711]
Weight loss	8.09 (7) [0.3251]	7.82 (7) [0.3489]	3.60 (3) [0.3083]
ADL increase	13.10 (7) [0.0697]	12.68 (7) [0.0802]	2.21 (3) [0.5306]
UTI	5.98 (7) [0.5422]	5.95 (7) [0.5450]	0.97 (3) [0.8081]

Notes: Each cell reports the Wald χ^2 statistic for the joint null that all pre-treatment event-time coefficients equal zero, followed by degrees of freedom in parentheses and the p-value in brackets.
The 2 Year Full Pre-Window tests $\tau = -8$ through $\tau = -1$, with $\tau = -1$ as the omitted reference period.
The ownership-change-quarter-excluded specifications omit $\tau = 0$ because the quarter of ownership change may combine pre- and post-transfer care, assessment, and documentation.
The 1 Year Window tests $\tau = -4$ through $\tau = -1$, with $\tau = -1$ as the omitted reference period.
Sample sizes by outcome: 2 Year Full Pre-Window [Catheter=188,762; Antipsychotic=201,403; Hypnotics=150,667; Pressure injuries=140,715; Falls=197,246; Weight loss=198,690; ADL increase=200,578; UTI=196,502].
Sample sizes by outcome: 2 Year, with Donut [Catheter=187,542; Antipsychotic=200,132; Hypnotics=149,711; Pressure injuries=139,764; Falls=195,998; Weight loss=197,438; ADL increase=199,311; UTI=195,249].
Sample sizes by outcome: 1 Year, with Donut [Catheter=187,542; Antipsychotic=200,132; Hypnotics=149,711; Pressure injuries=139,764; Falls=195,998; Weight loss=197,438; ADL increase=199,311; UTI=195,249].
All specifications include facility and quarter fixed effects and covariates: government + non_profit + chain + beds + occupancy_rate + pct_medicare + pct_medicaid + cm_q_state_2 + cm_q_state_3 + cm_q_state_4.

3.8.2 Additional Analyses

We examine heterogeneity in the effects of ownership change along three dimensions, in each case by estimating the specifications above on subsamples rather than by interacting treatment with facility characteristics. Sample splits are preferred here because they allow the fixed effects and control coefficients to differ across groups rather than imposing a common structure with a shifted intercept.

First, we estimate staffing models separately for the pre-pandemic period, January 2017 through December 2019, and the pandemic period, April 2020 through June 2024. Labor market conditions for nursing staff changed substantially during the COVID-19 pandemic, including heightened turnover, changes in wages, and staffing shortages. Estimating separately by period allows us to assess whether ownership changes had differential effects on staffing before and during this structural shift in the nursing labor market.

Second, we split facilities by baseline chain affiliation. Chain-affiliated facilities may have access to internal labor markets, standardized management practices, and administrative support that independent facilities lack, which could allow them to absorb an ownership transition with smaller staffing adjustments. Because chain status changes for a substantial share of facilities around their own transaction, we classify facilities using the baseline measure defined in Section 3 rather than the contemporaneous indicator.

Question: An alternative is to classify each facility by its own pre-event chain status over $\tau \in [-12, -4]$ rather than a common baseline month. This is arguably better matched to the transaction being studied; the two approaches disagree for a meaningful share of facilities and the choice should be settled before the results are interpreted.

Third, we examine whether staffing and business-model responses vary with the facility's baseline occupancy. Facilities operating with substantial unused capacity have a margin available to an incoming owner that facilities near capacity do not, so the scope for census growth, and therefore for the denominator effect discussed above, differs systematically across the occupancy distribution. We classify treated facilities into bins based on average occupancy over the twelve to four months preceding the ownership change, and estimate a specification interacting the post indicator with these bins. Never-treated facilities are retained in the estimation sample, where they contribute to identification of the calendar fixed effects.

In all three analyses we estimate the same specifications as in equations 2 and 4, maintaining consistent controls, fixed effects, and clustering.

4 Results

4.1 Summary Statistics

Table 3 reports summary statistics for the monthly staffing sample. The final sample consists of approximately 7,806 unique skilled nursing facilities and 632,231 facility-month observations spanning January 2017 through June 2024. Of these facilities, 1589 experience an ownership change during the sample period.

Table 3: Summary Statistics

Variable	Mean	SD	N
Panel A: Staffing sample (facility–month)			
Staffing outcomes			
RN HPRD	0.432	0.309	554,045
LPN HPRD	0.802	0.326	554,045
CNA HPRD	2.118	0.530	554,045
Total HPRD	3.352	0.777	554,045
RN hours (monthly)	1,030	945	554,045
LPN hours (monthly)	2,031	1,382	554,045
CNA hours (monthly)	5,312	3,358	554,045
Total hours (monthly)	8,373	5,103	554,045
Facility characteristics			
Non-profit	0.191	0.393	554,045
Chain affiliation (baseline)	0.572	0.495	554,045
Beds	107.5	56.4	554,045
Occupancy rate (%)	77.2	15.3	554,045
% Medicare	13.5	11.4	554,045
% Medicaid	60.1	20.8	554,045
Average length of stay (days)	160.9	157.9	544,030
Acuity quartile 2	0.230	0.421	554,045
Acuity quartile 3	0.233	0.423	554,045
Acuity quartile 4	0.240	0.427	554,045
Panel B: Quality sample (facility–quarter)			
Quality measures			
Catheter use	1.610	2.135	173,838
Anti-psychotic medication use	14.447	9.371	172,519
Anti-anxiety/hypnotic medication use	20.238	10.319	173,036
Pressure injuries	7.922	5.310	122,915
Falls with major injury	3.341	2.831	176,296
Weight loss	6.300	4.886	171,896
Decline in physical functioning	14.809	8.797	165,126
Urinary tract infections	2.511	3.250	175,551
Staffing and facility characteristics			
RN HPRD	0.429	0.303	183,977
LPN HPRD	0.800	0.320	183,977
CNA HPRD	2.113	0.518	183,977
Total HPRD	3.342	0.762	183,977
Non-profit	0.190	0.393	182,894
Chain affiliation (baseline)	0.572	0.495	183,977
Beds	107.8	56.3	183,977
Occupancy rate (%)	77.3	15.2	180,689
% Medicare	13.9	12.7	181,263
% Medicaid	60.1	20.9	164,327
Acuity quartile 2	0.248	0.432	183,977
Acuity quartile 3	0.251	0.434	183,977
Acuity quartile 4	0.259	0.438	183,977

Notes: N is the number of non-missing observations for each variable. HPRD denotes hours per resident day; monthly hours are the HPRD numerator. Chain affiliation is measured at the start of each facility's panel and held fixed. Acuity quartiles are within-state, within-period case-mix quartiles, with quartile 1 omitted. For every quality measure, lower values indicate better measured quality.

Panel A: 6,867 facilities (1,413 with an ownership change), 2017/01–2024/06. Panel B: 6,854 facilities (1,413 with an ownership change), 2017Q1–2024Q2.

Both samples exclude hospital-based facilities, facilities outside the 48 contiguous states, and any facility

4.2 Staffing Responses to Ownership Change

Figure 1 presents event study estimates of the effect of ownership changes on nursing staffing outcomes (i.e., Equation 2). Following an ownership transition, staffing declines persist for many periods, with the largest and most precise effects occurring for registered nurses (RN), certified nursing assistants (CNA), and total nursing hours per resident day.

RN staffing declines immediately following ownership changes and remains far below pre-transition levels throughout the post-treatment period. CNA staffing exhibits a similar but slightly smaller decline, while total nursing hours mirror the combined reductions across staff types. In contrast, LPN staffing shows limited and statistically insignificant changes over time. These dynamic patterns indicate that ownership changes are followed by sustained reductions in labor inputs, rather than short-run staffing adjustments that quickly reverse.

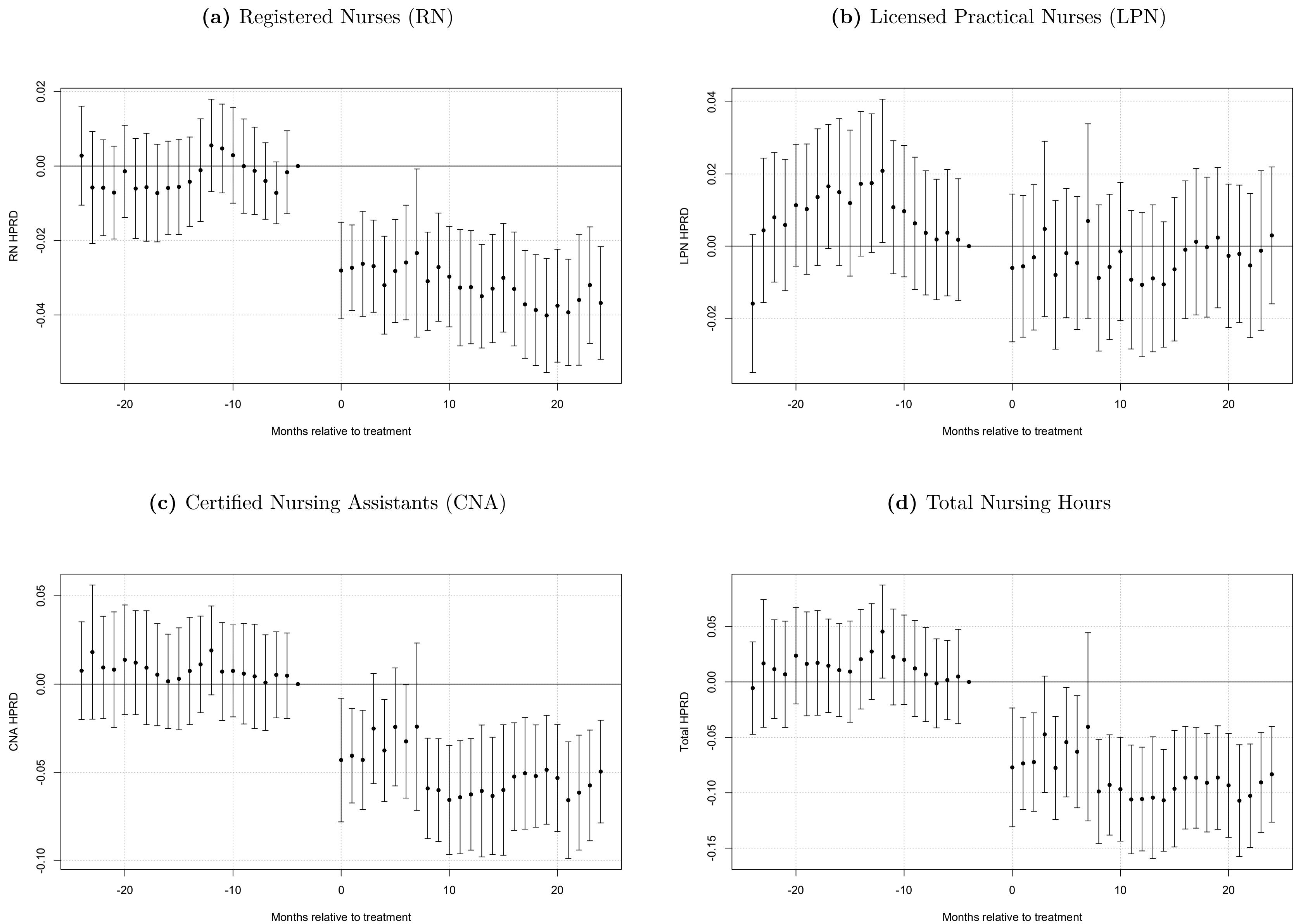


Figure 1: Dynamic Effects of Ownership Change on Nursing Staffing. Event study estimates of the effect of a change in ownership on nursing staff levels in hours per resident day (HPRD). Points show coefficient estimates relative to the reference period $\tau = -4$; vertical bars denote 95% confidence intervals. The three months prior to the ownership change ($\tau = -3, -2, -1$) are excluded. All specifications include control variables and facility and month fixed effects.

To summarize these dynamic effects with a single average treatment estimate, Table **??** reports two-way fixed effects estimates of the ownership change effect (i.e., Equation 4). Panel A reports results in levels, while Panel B reports logged results. Consistent with the event-study evidence, the average post-transition estimates indicate sustained reductions in nursing labor inputs. RN staffing declines by approximately 0.03 hours per resident day (roughly 9 percent), CNA staffing falls by about 0.05 hours (approximately 3 percent), and total staffing decreases by 0.08 hours per resident day (roughly 3 percent). LPN staffing shows no statistically significant change.

Table 4: Effect of Ownership Change on Nursing Staffing

Outcome	RN	LPN	CNA	Total
HPRD	-0.0368^{***}	-0.0039	-0.0532^{***}	-0.0940^{***}
	(0.0055)	(0.0062)	(0.0096)	(0.0134)
Log(HPRD)	-0.1174^{***}	-0.0022	-0.0270^{***}	-0.0293^{***}
	(0.0168)	(0.0095)	(0.0049)	(0.0041)
Hours	-80.0180^{***}	68.4756^{***}	64.8807^{*}	53.3383
	(15.3077)	(17.8010)	(34.9039)	(49.9070)
Log(hours)	-0.0767^{***}	0.0384^{***}	0.0133^{*}	0.0110^{*}
	(0.0173)	(0.0103)	(0.0068)	(0.0060)

Notes: Each cell reports the coefficient on *post*, with standard errors in parentheses. All specifications include facility and calendar-period fixed effects and control for the number of certified beds. Standard errors are two-way clustered by facility and calendar period. The sample excludes facilities government-owned at any point during the study period.
HPRD is hours per resident day; hours are the HPRD numerator, reported to test whether the HPRD decline is mechanically driven by the occupancy-rate increase in the denominator.
The anticipation window ($\tau = -3, -2, -1$) is excluded.
Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 5: Effect of Ownership Change on Business-Model Outcomes

Outcome	Coefficient (SE)	Observations
Occupancy rate	2.6841***	550,330
	(0.3497)	
Spare capacity	−0.0263***	550,330
	(0.0035)	
Medicare share	1.6100***	550,330
	(0.2159)	
Medicaid share	−1.1710*	550,330
	(0.5952)	
Average length of stay	−13.3641***	540,346
	(4.3491)	
Case mix	0.0525***	457,842
	(0.0059)	

Notes: Each cell reports the coefficient on *post*, with standard errors in parentheses. All specifications include facility and calendar-period fixed effects and control for the number of certified beds. Standard errors are two-way clustered by facility and calendar period. The sample excludes facilities government-owned at any point during the study period.
Occupancy rate is residents as a share of available bed-days; spare capacity is unused certified beds as a share of certified beds. Payer shares are shares of patient days. The anticipation window ($\tau = -3, -2, -1$) is excluded.
Case mix is restricted to 2017/10–2023/12. CMS changed its case-mix methodology in October 2017 and January 2024; both breaks shift the level and the cross-sectional spread simultaneously across the whole sample, which facility and period fixed effects do not absorb.
Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

4.2.1 Heterogeneity Before and After the Onset of COVID-19

Table **??** examines whether these effects differ before and after the COVID-19 pandemic. The estimated reductions are generally smaller during the pandemic period for CNA and total HPRD, while RN reductions remain negative and statistically significant. This pattern suggests that pandemic-era labor market constraints and heightened regulatory scrutiny may have limited some staffing adjustments after ownership change.

Table 6: Effect of Ownership Change on Nursing Staffing: Pre-Pandemic vs. Pandemic

Outcome	RN	LPN	CNA	Total
Pre-pandemic (2017/01–2019/12) (N = 222,399 facility-months)				
HPRD	-0.0320^{***}	-0.0176^{*}	-0.0562^{***}	-0.1057^{***}
	(0.0068)	(0.0088)	(0.0143)	(0.0205)
Log(HPRD)	-0.0991^{***}	-0.0100	-0.0292^{***}	-0.0328^{***}
	(0.0253)	(0.0126)	(0.0071)	(0.0064)
Pandemic (2020/04–2024/06) (N = 312,471 facility-months)				
HPRD	-0.0279^{***}	-0.0062	-0.0546^{***}	-0.0887^{***}
	(0.0076)	(0.0085)	(0.0157)	(0.0216)
Log(HPRD)	-0.0959^{***}	-0.0059	-0.0244^{***}	-0.0269^{***}
	(0.0220)	(0.0114)	(0.0081)	(0.0064)

Notes: Each cell reports the coefficient on *post*, with standard errors in parentheses. All specifications include facility and calendar-month fixed effects and control for the number of certified beds. Standard errors are two-way clustered by facility and calendar month. The sample excludes facilities government-owned at any point during the study period. The anticipation window ($\tau = -3, -2, -1$) is excluded.
Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 7: Effect of Ownership Change on Nursing Staffing: Chain vs. Non-Chain Facilities

Outcome	RN	LPN	CNA	Total
Chain (baseline) (N = 314,227 facility-months)				
HPRD	-0.0349^{***}	0.0063	-0.0259^{**}	-0.0545^{***}
	(0.0067)	(0.0074)	(0.0110)	(0.0153)
Log(HPRD)	-0.1302^{***}	0.0062	-0.0146^{**}	-0.0189^{***}
	(0.0199)	(0.0116)	(0.0056)	(0.0048)
Non-chain (baseline) (N = 236,107 facility-months)				
HPRD	-0.0360^{***}	-0.0213^{*}	-0.0980^{***}	-0.1553^{***}
	(0.0097)	(0.0109)	(0.0181)	(0.0254)
Log(HPRD)	-0.0803^{***}	-0.0161	-0.0470^{***}	-0.0449^{***}
	(0.0304)	(0.0165)	(0.0092)	(0.0076)

Notes: Each cell reports the coefficient on *post*, with standard errors in parentheses. All specifications include facility and calendar-month fixed effects and control for the number of certified beds. Standard errors are two-way clustered by facility and calendar month. The sample excludes facilities government-owned at any point during the study period. The anticipation window ($\tau = -3, -2, -1$) is excluded.
Chain status is each facility's baseline classification (*chain_at_start*): chain status in January 2017, falling back to the facility's own earliest observed value if absent from the panel that month. Facilities with no available chain classification are excluded from both panels.
Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

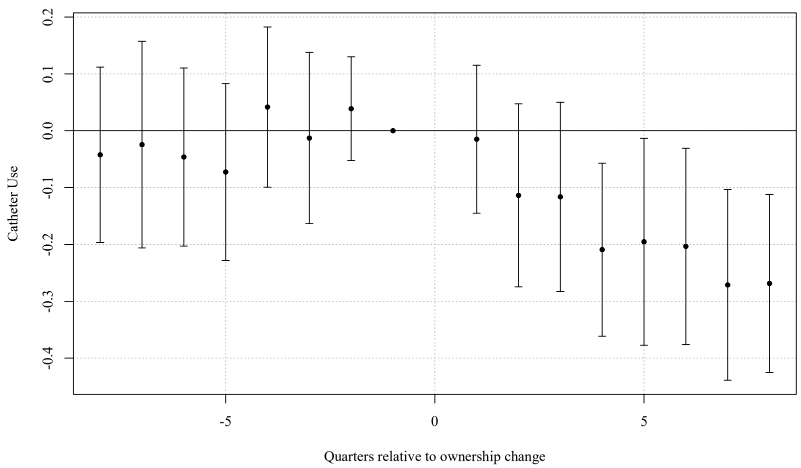
Together, the staffing results show that ownership changes are followed by sustained reductions in measured nursing labor inputs. The declines are concentrated among registered nurses, certified nursing assistants, and total nursing hours per resident day, while licensed practical nurse staffing remains comparatively stable. These patterns suggest that ownership changes are not merely legal or administrative events, but events of operational restructuring in which new owners adjust the mix and intensity of labor used in care production.

The welfare implications of these input reductions are ambiguous without examining quality outcomes. If staffing reductions primarily reflect harmful cost-cutting, one would expect worsening in resident outcomes, particularly measures that depend heavily on direct staff attention. If, instead, ownership changes disrupt old routines and move facilities closer to their feasible production frontier, reductions in labor inputs may occur alongside improvements in quality measures that are sensitive to managerial routines, monitoring, and care protocols.

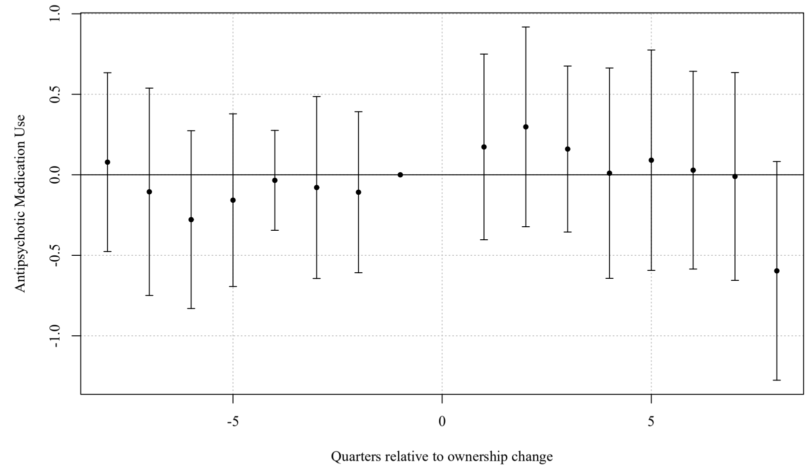
4.3 Quality Responses to Ownership Change

Figure 2 presents event-study estimates for routine-sensitive quality measures. These outcomes are intended to capture changes in care that may respond to changes in oversight, monitoring, documentation, and clinical routines. The estimates show whether ownership changes are followed by changes in quality measures that may be especially responsive to organizational re-optimization.

(a) Catheter Use



(b) Anti-Psychotic Medication Use



(c) Anti-Anxiety or Hypnotic Medication Use

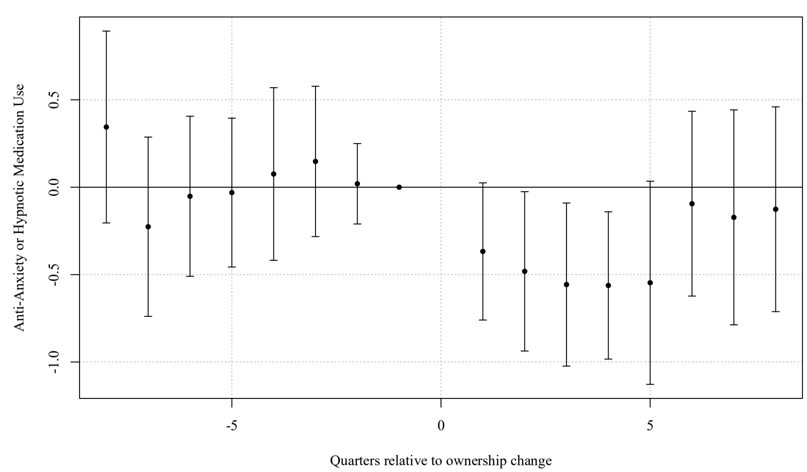
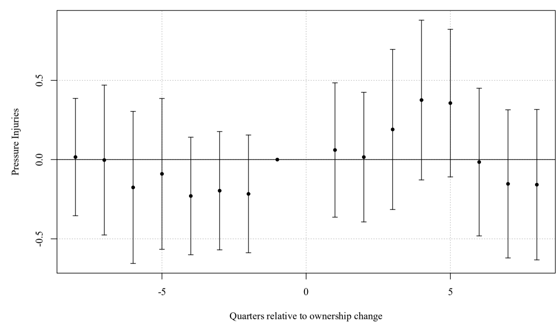


Figure 2: Dynamic Effects of Ownership Change on Labor Saving Mechanisms. Event-study estimates of the effect of a change in ownership on labor saving mechanism measures. Points show coefficient estimates relative to the reference period $\tau = -1$; vertical bars denote 95% confidence intervals. The ownership-change quarter ($\tau = 0$) is excluded from the estimation sample. All specifications include control variables and facility and quarter fixed effects.

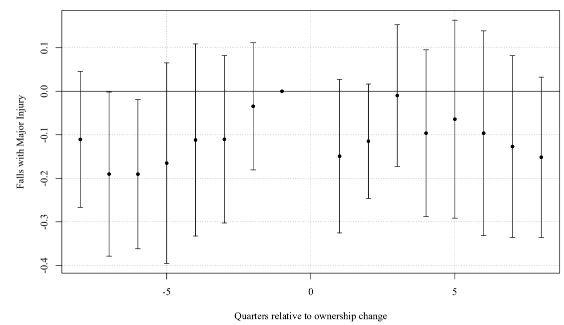
The results shown in figure 2 provide mixed but generally supportive evidence that ownership changes alter clinical and managerial routines. Catheter use declines following ownership change, and anti-anxiety or hypnotic medication use also falls relative to the reference period. Antipsychotic medication use declines minimally but is not statistically significant. Because lower values indicate better measured quality for all quality outcomes, these patterns suggest that ownership changes are not followed solely by reductions in measured inputs; some quality measures that capture labor saving mechanisms improve after control changes.

Figure 3 presents event-study estimates for staffing-intensive resident outcomes. These outcomes are more directly tied to hands-on care intensity. If ownership changes primarily reduced inputs in ways that harmed residents, these measures would be the most likely to worsen.

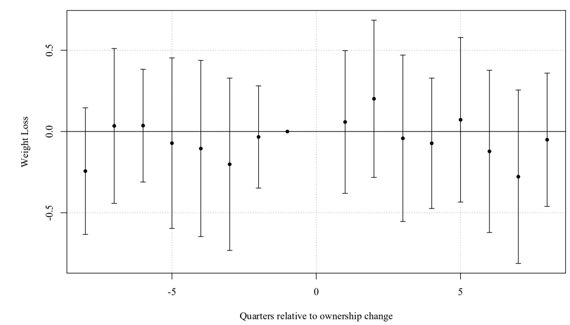
(a) Pressure Injuries



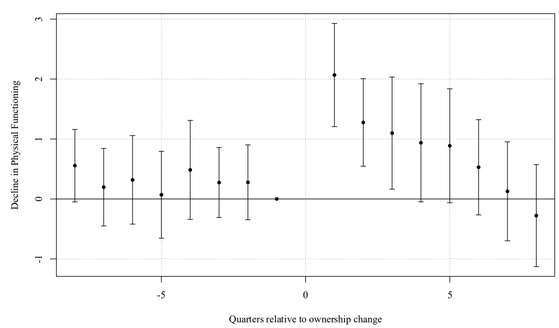
(b) Falls with Major Injury



(c) Weight Loss



(d) Decline in Physical Functioning



(e) Urinary Tract Infections

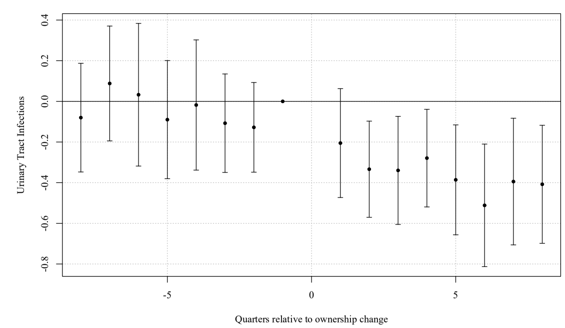


Figure 3: Dynamic Effects of Ownership Change on Resident Outcome Measures. Event-study estimates of the effect of a change in ownership on resident outcome measures. Points show coefficient estimates relative to the reference period $\tau = -1$; vertical bars denote 95% confidence intervals. The ownership-change quarter ($\tau = 0$) is excluded from the estimation sample. All specifications include control variables and facility and quarter fixed effects.

The resident outcome measures show less evidence of broad worsening. Pressure injuries, falls with major injury, weight loss, and decline in physical functioning do not change in a statistically significant manner. Urinary tract infections do show modest declines. Taken together, these results provide little evidence that staffing reductions, through ownership changes, translate into widespread worsening of the selected resident outcome measures over the observed window.

Table **??** summarizes the quality responses using average post-change TWFE specifications.

Table 8: Effect of Ownership Change on Quality Measures

Outcome	(1)	(2)	Observations
Panel A: Long-stay labor-saving mechanism measures			
Catheter use	-0.1695^{***}	-0.1711^{***}	172,671
	(0.0488)	(0.0488)	
Anti-psychotic medication use	-0.3362	-0.3543	171,358
	(0.2218)	(0.2231)	
Anti-anxiety/hypnotic medication use	-0.2321	-0.2118	171,854
	(0.2019)	(0.2027)	
Panel B: Long-stay resident outcome measures			
Pressure injuries	0.2586	0.2825	121,962
	(0.1762)	(0.1783)	
Falls with major injury	0.0125	0.0138	175,123
	(0.0636)	(0.0644)	
Weight loss	0.2042^{*}	0.2080^{*}	170,733
	(0.1040)	(0.1034)	
Decline in physical functioning	0.3240	0.3501	163,983
	(0.2695)	(0.2782)	
Urinary tract infections	-0.3279^{***}	-0.3181^{***}	174,381
	(0.0765)	(0.0767)	
Panel C: Short-stay measures			
New antipsychotic medication	-0.0187	-0.0219	134,020
	(0.0432)	(0.0433)	
Improved function	-0.0569	-0.0880	88,792
	(0.4888)	(0.4889)	

Notes: Each cell reports the coefficient on *post*, with standard errors in parentheses. All specifications include facility and calendar-period fixed effects and control for the number of certified beds. Standard errors are two-way clustered by facility and calendar period. The sample excludes facilities government-owned at any point during the study period.

Column (1) is the baseline specification. Column (2) adds RN, LPN, and CNA hours per resident day as controls. Because staffing is itself affected by ownership change, column (2) is not a preferred estimate of the total effect; it is reported to show whether the quality response is attenuated after conditioning on measured staffing inputs.

Long-stay measures (Panels A-B) and short-stay measures (Panel C) are constructed from different resident populations and are not directly comparable to one another. For every measure, lower values indicate better measured quality.

Pressure injuries is estimated on 2018–2023 only; improved function is estimated on 2017–2022 only. Both measures show near-complete absence outside these windows and are trimmed to the years where they are actually reported rather than treated as full-panel outcomes. Vaccination measures are excluded from this table.

The transition quarter ($\tau = 0$) is excluded. Observations are reported for column (1); column (2) drops facility-quarters with missing staffing.

Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

The estimates are similar with and without staffing controls. Because staffing is itself affected by ownership change, the staffing-controlled specifications are not interpreted as preferred total effects. Instead, their similarity to the baseline estimates suggests that the measured quality responses are not mechanically explained by contemporaneous changes in RN, LPN, and CNA hours per resident day.

4.4 Robustness

This section reports robustness checks for our staffing results, which are the main input-side estimates in our paper. These checks are designed to assess the validity of our identifying assumptions and the sensitivity of our results to alternative model choices. Across all exercises, the estimated effects of ownership change on staffing remain stable in sign, magnitude, and statistical significance. The supporting tables and figures are reported in the appendix.

4.4.1 Alternative Comparison Groups

Our main event-study specification is estimated using two-way fixed effects (TWFE) with staggered treatment timing. In this setting, TWFE can implicitly use already-treated facilities as controls for cohorts treated later, which may distort event-study estimates when treatment effects are dynamic or heterogeneous across cohorts. To assess sensitivity to this issue, we re-estimate the event study using a stacked difference-in-differences design.

In the stacked approach, we construct cohort-specific samples for each treatment month g. For a given cohort, treated facilities are those with an ownership change in month g, and the control group consists only of facilities that are never treated or not-yet-treated relative to g (i.e., with treatment dates after g). We restrict each cohort-specific sample to a symmetric event window around g and then stack these cohort datasets into a single analysis file. Estimation includes the same covariates as the baseline specification and fixed effects that absorb facility-level time-invariant heterogeneity and common calendar shocks; standard errors are clustered at the facility level.

Appendix Figures 5a through 5d show that the stacked event-study estimates closely track the baseline TWFE event-study results: the post-transition dynamics and the relative magnitudes across staffing types remain qualitatively unchanged. In addition, Appendix Table 13 reports joint Wald tests of pre-treatment coefficients in the stacked specification (under the donut design used in the baseline), which are consistent with flat pre-trends in the remaining pre-period. Together, these results suggest that the main findings are not driven by treated-as-control comparisons that can arise in TWFE under staggered adoption.

4.4.2 Event Window and "Donut"

As we have previously discussed, because of changes to staffing in anticipation of a formal ownership change, the assumptions of parallel trends is violated. As a result of this, in our baseline specifications we "donut" equation 2 and equation 4 by removing $\tau \in \{-3, -2, -1\}$. To test the validity of removing this set of pre-treatment coefficients, we estimate equations 2 and 4 with different "donut" sets and show that the estimated post-treatment effects remain largely unchanged. Row 6 of Table 12 reports estimates of equation 4 with a wider anticipatory period excluded from the sample. The results from row 6 show no differences in values or statistical significance. Similarly, row 7 reports estimates under a smaller anticipatory window that is excluded from the data, and estimates remain similar in magnitude and statistical significance.

4.4.3 Gap Size

Because staffing data is occasionally missing or reported with gaps, we examine whether our results are sensitive to the length of time between consecutive observations. Restricting the sample to facilities with smaller reporting gap shows that estimates are similar to those in the full sample, indicating that data gaps do not drive our findings. Row 2 of Table 12 reports our results for equation 4 with the restriction that facilities with a reporting gap of larger than 6 are excluded from our sample. Compared to the baseline (row 1) this result is identical across staff types and statistical significance. The exact same result is true for row 3 which reports estimates for equation 4 where facilities with a reporting gap of greater than 3 are removed. Row 4 and 5 report estimates where gap greater than 1 or equal to 0 respectively, are removed. Similarly, we see no differences in these results with respect to the baseline.

5 Discussion

This paper studies ownership changes in nursing homes as moments of organizational disruption in care production. Our results show that ownership transitions are followed by sustained reductions in measured nursing labor inputs, especially RN hours, CNA hours, and total nursing hours per resident day. At the same time, we find that some selected quality measures improve: catheter use and anti-anxiety or hypnotic medication use decline, and urinary tract infections also fall. Other resident outcome measures, including pressure injuries, falls with major injury, weight loss, and decline in activities of daily living, show little evidence of broad worsening. Taken together, these patterns suggest that ownership changes do more than alter the legal identity of the owner. They appear to change how facilities organize production.

The combination of lower staffing inputs and selected quality improvements is consistent with the idea that some facilities operate inside their feasible production frontier before undergoing ownership change. New owners do not necessarily possess a superior care technology. Instead, the ownership transition may disrupt old routines, alter managerial oversight, change medication-management practices, or reduce organizational slack. If facilities, under previous ownership, relied on inefficient routines, then new owners could reduce some inputs while maintaining or improving selected measured quality outcomes. This is consistent with the idea that ownership change may not be a mechanism for cost-containment, but rather a way for new owners to re-optimize production inputs.

5.1 Interpreting Staffing Reductions

The results of our staffing specifications indicate that both the level and mix of nurse staffing adjust following a change in ownership. The declines are concentrated among RNs and CNAs, while LPN staffing remains comparatively stable. In a regulated sector with limited short-run pricing flexibility, staffing is one of the most immediately adjustable cost-saving mechanisms available to new owners.

The relative stability of LPN staffing is informative because LPNs occupy an intermediate position in nursing home production. RNs perform higher-skill clinical and supervisory functions, while CNAs provide much of the hands-on daily care (Lin, 2014). LPNs can perform routine clinical tasks and help maintain day-to-day coverage, making them a potential margin of adjustment when facilities reduce RN and CNA hours. The absence of a comparable decline in LPN staffing is therefore consistent with compositional adjustment rather than uniform staffing cuts across all nurse types.

An important question that can be raised is whether the estimated staffing reductions are economically large. Benchmarking the estimates against baseline staffing levels helps provide context. In our sample, mean staffing levels are 0.427 RN HPRD, 0.796 LPN HPRD, 2.100 CNA HPRD, and 3.324 total HPRD. In our average post-treatment effect specification, ownership change reduces RN staffing by approximately 0.03 HPRD, CNA staffing by approximately 0.05 HPRD, and total staffing by approximately 0.08 HPRD. Relative to baseline staffing levels, these estimates imply declines of roughly 7 to 9 percent for RN staffing and about 3 percent for CNA and total staffing. Expressed in minutes, the total staffing reduction is approximately five fewer nursing minutes per resident-day. These changes may be modest and should not, by themselves, be seen as evidence of substantial welfare losses. However, staffing is a core input into care production, so even small reductions may matter in facilities operating close to staffing constraints or for residents requiring more intensive assistance.

5.2 Quality Responses and the Reorganization of Care

The results of our regressions on quality help distinguish between alternative interpretations of the staffing reductions. If ownership changes primarily reflected quality-reducing cost containment, we would expect staffing-intensive resident outcomes to become worse after the ownership change. The evidence does not show broad worsening in the selected resident outcome measures. Pressure injuries, falls with major injury, weight loss, and decline in physical functioning are not statistically distinguishable from zero in the average post-change specifications. Urinary tract infections decline.

At the same time, we find that labor saving mechanism measures improve. Catheter use falls, and anti-anxiety or hypnotic medication use declines after ownership change. Anti-psychotic medication use also declines moderately, though the average post-change estimate is not statistically significant. These measures may respond to changes in care routines, medication management, monitoring, documentation, and managerial oversight. Their improvement suggests that ownership changes may lead facilities to revise care processes rather than simply reduce labor inputs.

This pattern is consistent with incumbent nursing home owners operating their facilities inside their feasible production frontier. A facility can be inside its feasible production frontier if existing routines persist even when better combinations of inputs and outcomes are attainable. Ownership change may interrupt those routines by introducing new managers, new monitoring systems, different performance targets, or greater willingness to change staffing and care practices. Under this interpretation, lower RN and CNA hours do not necessarily imply a proportional decline in care quality. Instead, they may reflect a reorganization of production in which some inefficient practices are reduced while selected quality measures improve.

5.3 Heterogeneity and Constraints

The heterogeneity results of our additional analyses provide insight into the constraints facing new owners. Staffing reductions are smaller during the COVID-19 period for several outcomes, suggesting that pandemic-era labor market tightness, heightened scrutiny, and operational disruption may have limited owners' ability to adjust staffing. When labor markets are tight or regulatory attention is elevated, the feasible set of staffing reductions may narrow.

Differences by organizational structure may also matter. Chain-affiliated facilities may have access to internal labor markets, standardized management practices, and administrative support that allow them to absorb ownership changes with smaller staffing disruptions. Independent facilities may face tighter financial or managerial constraints, making staffing a more immediate adjustment margin. These patterns are consistent with the broader interpretation that ownership changes operate through internal production decisions and that the scope for adjustment depends on capacity and constraints.

5.4 Limitations

Several limitations should be considered when interpreting our results. First, despite our two-step approach to identifying ownership transitions, the timing of ownership change may still be measured with error. The recorded ownership change date may not capture earlier contract negotiations, transition planning, or operational changes that occur before the formal closing date. Our preferred staffing specification addresses this concern by excluding the immediate pre-transition months, and our preferred quality specification excludes the ownership-change quarter, but some timing error may remain.

Second, ownership changes are not randomly assigned. The empirical design helps address time-invariant facility differences and common shocks, but ownership changes may still coincide with unobserved facility-level shocks such as financial distress, management turnover, or local labor market changes. These shocks could affect both staffing and quality.

Third, the staffing and quality measures are imperfect. PBJ staffing data may contain reporting gaps or measurement error, although robustness checks suggest that reporting gaps do not drive the main staffing results. CMS quality measures are also limited. They capture selected process and resident outcome measures but do not fully measure resident welfare, resident experience, deficiencies, hospitalizations, mortality, or other aspects of quality.

Finally, our results should not be interpreted as proving that ownership changes improve welfare. The evidence shows that ownership changes reduce selected labor inputs while some quality measures improve and others do not worsen. This pattern is consistent with routine replacement and inside-frontier inefficiency, but it is also possible that unmeasured aspects of care worsen or that effects differ across resident types and facilities. Future work should examine whether these ownership changes affect broader outcomes and whether the observed input reductions are concentrated in facilities with greater pre-transition slack.

6 Conclusion

Ownership changes are common in U.S. nursing homes, yet their implications for care production remain poorly understood. This paper studies ownership transitions as moments of organizational disruption and reoptimization. Using CMS administrative panel data from 2017 through 2024, we identify ownership transitions that reflect changes in economic control and estimate their effects on staffing inputs and selected quality outcomes. We find that ownership changes are followed by sustained reductions in RN staffing, CNA staffing, and total nursing hours per resident day, while LPN staffing remains comparatively stable. These staffing responses indicate that ownership transitions alter the internal organization of care production rather than merely changing the legal identity of the facility owner.

Our results focused on quality complicate a simple cost-cutting interpretation. If ownership changes reduced staffing in ways that broadly harmed residents, we would expect worsening in staffing-intensive resident outcomes. Instead, labor saving mechanisms improve, including catheter use and anti-anxiety or hypnotic medication use, while urinary tract infections also decline. Other resident outcome measures, including pressure injuries, falls with major injury, weight loss, and decline in physical functioning, show little evidence of broad worsening over the observed post-transition window. These patterns suggest that input reductions after ownership change do not translate mechanically into measured quality declines.

Our findings are consistent with ownership transitions revealing inside-frontier inefficiency in nursing home production. New owners do not necessarily possess a superior care technology, but they may be more willing to revise old routines, alter staffing composition, reduce organizational slack, and change medication-management or monitoring practices. In this sense, ownership change may operate as a disruption that shifts facilities toward a more efficient input-quality combination. At the same time, the welfare implications remain ambiguous. Our analysis does not directly estimate a production frontier, and the quality measures used do not capture all dimensions of resident welfare. Unmeasured outcomes, resident experience, deficiencies, hospitalizations, and mortality may respond differently.

This paper contributes to the literature by shifting the focus from ownership changes as solely market-structure events to ownership changes as shocks to the internal organization of care. In nursing homes, where prices are heavily regulated and staffing is a central input, changes in control can reshape production even when the facility remains in the same market and continues serving the same local population. Future work should examine which types of ownership changes generate efficiency-enhancing reorganization, which produce quality-reducing cost containment, and whether the observed input reductions are concentrated in facilities with greater pre-transition organizational slack.

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Tables

Table 9: Examples of Discordant Ownership Records Across HCRIS and NHC

Example	Classification	Period	NHC owner names and shares
Panel A: HCRIS-reported ownership changes not verified as substantive transfers in NHC records			
A	HCRIS only	$t = 1$	Karen Gail Miller Irrev. Tr. FBO Zane Miller (50%); The Gail Miller GST (50%)
A	HCRIS only	$t = 2$	Karen Gail Miller Irrev. Tr. FBO Zane Miller (20%); The Gail Miller GST (20%); Karen Miller-related trust entities (60%)
B	HCRIS only	$t = 1$	Vetter Eldora (50%); Vetter Jack (50%)
B	HCRIS only	$t = 2$	Vetter Senior Living (100%)
C	HCRIS only	$t = 1$	Abri Health Services (50%); The Senior Care Liquidation Co., Alan Halperin Trustee (50%)
C	HCRIS only	$t = 2$	Abri Care (100%)
D	HCRIS only	$t = 1$	Concordia Lutheran Ministries of Pittsburgh (100%)
D	HCRIS only	$t = 2$	Concordia Care Network (100%)
Panel B: NHC-identified ownership changes not reported as ownership changes in HCRIS			
E	NHC only	$t = 1$	Destefane Richard (100%)
E	NHC only	$t = 2$	Richard J. Destefane Revocable Living Trust (100%)
F	NHC only	$t = 1$	D'Angelo Cara (25%); Durham Deborah (25%); Horace D'Angelo Jr. Operating Entities Dated 01/01/2012 (25%); Tackett Kimberly (25%)
F	NHC only	$t = 2$	Benoit (9.1%); Calumet South (9.1%); Drake Louis Enterprise (9.1%); Fairhome Tr. UAD 12312012 (9.1%); Fairhome UAD 12312012 (9.1%); GZLT (9.1%); Hartman David (9.1%); Hartman Mark (9.1%); Krupp Ari (9.1%); Senderowicz Yossi (9.1%); Willow Delta (9.1%)
G	NHC only	$t = 1$	CMC II Investors (33.3%); Conte Joseph (33.3%); FC LV Holdco (33.3%)
G	NHC only	$t = 2$	Qazi Mohammad (100%)
H	NHC only	$t = 1$	Berkshire Healthcare Systems (67%); Integritus Healthcare (25%); Berkshire Healthcare System (3%)
H	NHC only	$t = 2$	Integritus Healthcare Management Services (100%)
I	NHC only	$t = 1$	DES C 2009 GRAT (100%)
I	NHC only	$t = 2$	DES 2009 Family (100%)

Notes: This table presents examples of discordant ownership-change classifications across HCRIS and the NHC Archive in a panel-style format. Panel A reports cases in which HCRIS reports an ownership change, but the NHC ownership records do not show a clear transfer to a new unrelated controlling owner. Panel B reports cases in which NHC ownership records show large changes in listed owners or ownership shares, but no corresponding HCRIS ownership-change flag is observed. Owner names and shares are taken from the NHC Archive. The labels $t = 1$ and $t = 2$ refer to the ownership records immediately before and after the relevant ownership-record change.

Table 10: Ownership-Change Classification and Sample Construction

Panel A. Cross-classification of ownership-change indicators

	NHC ownership records		
HCRIS ownership-change status	0	1	2+
0	XXX	XXX	XXX
1	XXX	XXX	XXX
2+ changes	XXX	XXX	XXX

Panel B. Construction of analytic ownership-change sample

Sample restriction / classification step	Facilities
Matched NHC–HCRIS facilities after initial ownership-record cleaning	XXX
Exclude hospital-based nursing homes	XXX
Exclude facilities outside the 48 contiguous U.S. states	XXX
Exclude facilities with unresolved or missing provider identifiers	XXX
Exclude facilities with multiple ownership changes in either source	XXX
Keep facilities with no ownership change in either source	XXX
Keep facilities with exactly one ownership change in both sources	XXX
Restrict matched ownership-change dates to within six months of one another	XXX
Final analytic ownership-change sample	XXX

Notes: Panel A reports the cross-classification of facility-level ownership-change indicators across HCRIS and the NHC Archive after common ownership-record cleaning and pruning. NHC ownership changes are classified using monthly ownership records and are intended to capture substantive transfers of economic control. Panel B reports the stepwise construction of the analytic ownership-change sample. The final sample includes facilities with no ownership change in either source and facilities with exactly one ownership change in both sources, where the HCRIS and NHC ownership-change dates occur within six months of one another. Counts are at the facility level.

Table 11: TWFE DiD Estimates of *post* on Staffing Outcomes (Stacked Sample, Baseline Donut)

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
HPRD	-0.024^{***}	-0.009	-0.051^{***}	-0.084^{***}
	(0.004)	(0.005)	(0.009)	(0.012)
Log(HPRD)	-0.070^{***}	-0.005	-0.027^{***}	-0.027^{***}
	(0.014)	(0.008)	(0.004)	(0.004)

Notes: Each cell reports the coefficient on *post* with standard errors in parentheses. The stacked sample includes never-treated and not-yet-treated controls for each cohort; the donut excludes $\tau \in \{-3, -2, -1\}$. Sample sizes: $N_{\text{HPRD}} = 24,197,338$; $N_{\text{Log}} = 23,819,333$.

Specifications include facility-by-cohort fixed effects (stack_id), calendar-month fixed effects, cohort fixed effects, and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.

Standard errors are clustered two ways by facility and calendar month.

Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 12: TWFE estimates of *post* on staffing levels (HPRD): sample-restriction and case-mix robustness.

	N	Outcomes			
		RN	**LPN**	**CNA**	**Total**
(1) Baseline (no anticipation)	550,334	-0.0368^{***}	-0.0039	-0.0532^{***}	-0.0940^{***}
		(0.0055)	(0.0062)	(0.0096)	(0.0134)
(2) Sample excludes *gap* > 6	550,175	-0.0368^{***}	-0.0040	-0.0532^{***}	-0.0941^{***}
		(0.0055)	(0.0062)	(0.0096)	(0.0134)
(3) Sample excludes *gap* > 3	549,918	-0.0368^{***}	-0.0040	-0.0532^{***}	-0.0940^{***}
		(0.0055)	(0.0061)	(0.0096)	(0.0134)
(4) Sample excludes *gap* > 1	546,319	-0.0370^{***}	-0.0037	-0.0533^{***}	-0.0940^{***}
		(0.0055)	(0.0061)	(0.0096)	(0.0134)
(5) Sample excludes *gap* > 0	546,319	-0.0370^{***}	-0.0037	-0.0533^{***}	-0.0940^{***}
		(0.0055)	(0.0061)	(0.0096)	(0.0134)
(6) Drop $t \in \{-4, -3, -2, -1\}$	549,052	-0.0365^{***}	-0.0037	-0.0530^{***}	-0.0932^{***}
		(0.0056)	(0.0063)	(0.0097)	(0.0136)
(7) Drop $t \in \{-2, -1\}$ only	551,637	-0.0370^{***}	-0.0040	-0.0530^{***}	-0.0941^{***}
		(0.0055)	(0.0061)	(0.0094)	(0.0132)
(8) + Case-mix: State Quartile	550,334	-0.0384^{***}	-0.0057	-0.0577^{***}	-0.1019^{***}
		(0.0055)	(0.0062)	(0.0095)	(0.0134)
(9) + Case-mix: State Decile	550,334	-0.0388^{***}	-0.0060	-0.0584^{***}	-0.1031^{***}
		(0.0055)	(0.0062)	(0.0095)	(0.0133)
(10) + Case-mix: National Quartile	550,334	-0.0388^{***}	-0.0056	-0.0581^{***}	-0.1026^{***}
		(0.0055)	(0.0062)	(0.0095)	(0.0134)
(11) + Case-mix: National Decile	550,334	-0.0395^{***}	-0.0059	-0.0589^{***}	-0.1043^{***}
		(0.0055)	(0.0062)	(0.0095)	(0.0134)

Notes: Each cell reports the *post* coefficient on staffing levels (HPRD), with two-way clustered standard errors (facility and calendar month) in parentheses.

Rows (1)–(7) hold the control set fixed at Spec A (post + number of certified beds, plus facility and calendar-month fixed effects) and vary the sample restriction or anticipation-window definition, as described in each row label. Row (1) is the project's standard sample: the anticipation window ($\tau = -3, -2, -1$) excluded, no additional gap restriction.

Rows (8)–(11) hold the sample fixed at row (1) and add one case-mix control set on top of Spec A per row – Spec A itself includes no case-mix control, so these rows are not directly comparable to rows (1)–(7). "State Quartile" is the project's default case-mix control elsewhere in the paper (see _setup.R's `preferred_case_mix_controls`). Quartile bins use dummies for bins 2–4; decile bins use dummies for bins 2–10 (bin 1 omitted as reference in both cases).

Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 13: Joint Wald Tests of Pre-Trends: Stacked Event Study (Levels)

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
2 Year Window with Donut	34.45 (20)	20.65 (20)	20.06 (20)	25.86 (20)
	[0.0233]	[0.4178]	[0.4539]	[0.1706]
1 Year Window with Donut	8.62 (8)	11.90 (8)	5.48 (8)	11.83 (8)
	[0.3752]	[0.1558]	[0.7048]	[0.1588]

Notes: Each cell reports the Wald χ^2 statistic for the joint null that all pre-treatment event-time coefficients equal zero, followed by degrees of freedom in parentheses and the p-value in brackets.

Tested windows and reference periods: 2 Year Window with Donut tests $\tau = -24$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$); 1 Year Window with Donut tests $\tau = -12$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$).

Sample sizes (stacked rows): 2 Year Window with Donut ($N = 24,197,338$); 1 Year Window with Donut ($N = 13,268,799$).

Stacked design: each treated cohort is compared only to never-treated and not-yet-treated facilities within its own event window; the donut excludes $\tau = -3, -2, -1$ (physically dropped from the estimation sample for treated units).

All specifications include facility-by-cohort fixed effects (stack_id), calendar-month fixed effects, cohort fixed effects, and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators. Standard errors are clustered by facility.

Figures

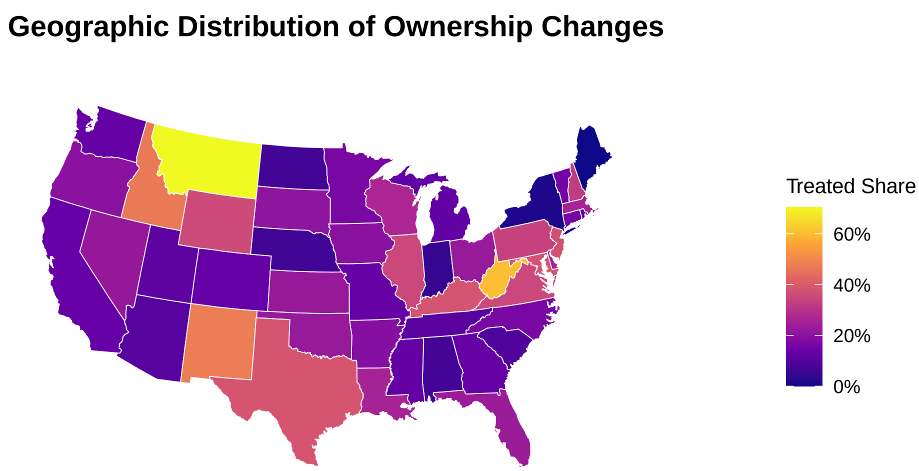

Figure 4: Geographic Distribution of Ownership Changes. The figure reports the share of nursing homes in each state that experienced an ownership change during the study period.

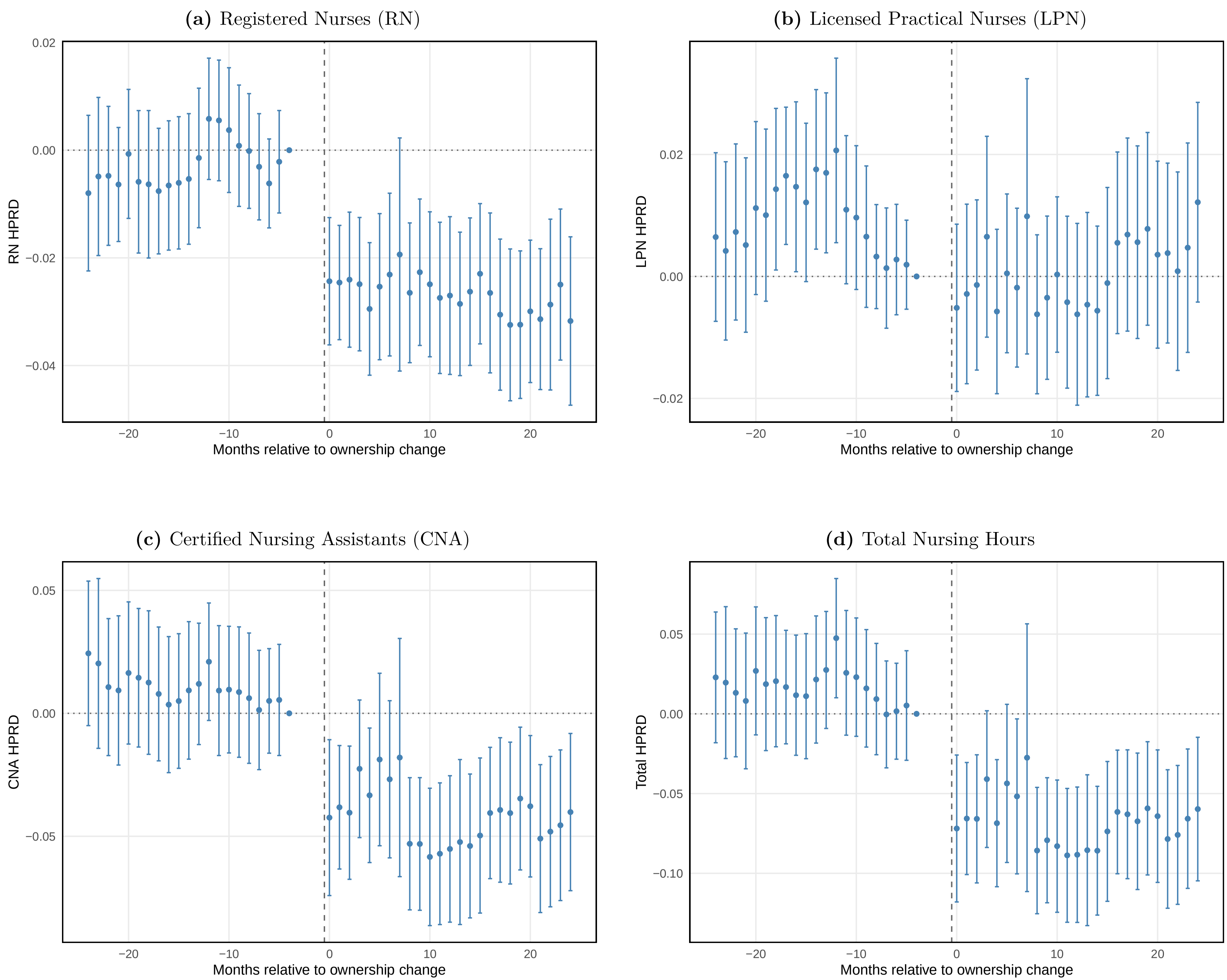

Figure 5: Stacked Event-Study Estimates of Ownership Change on Nursing Staffing. The stacked design compares each treated cohort only to never-treated and not-yet-treated facilities within the event window, preventing already-treated facilities from serving as controls. The donut excludes $\tau \in \{-3, -2, -1\}$; the reference period is $\tau = -4$.